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A Multi-dimensional Dynamic Measure for Detecting Local Speech Rate Variation

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1. Introduction

It is known that the local speech rate variations are often considered as cues to suggest the speaker's intention[1]. For example, when the speaker wants to emphasize something he/she may slow down his/her utterance and speaks clearly in some case. Then the listener acquires such information. On the other hand, our previous study[2] demonstrates that people who have no study experience on an unknown foreign language can perceive the different rate of the utterance. However, it is still not very clear that how a listener can get speech rate information from the utterance.

There have been attempts to estimate the speech rate by analyzing the acoustic properties. The average transition duration derived from the cepstral difference curve using English (TIMIT database) is found to have significant correlation with conventional measures such as phoneme rate [3]. In our previous study [2], we investigated the relation between dynamic measure of LPC cepstrum and speech rate using Chinese mandarin and also observed linear correlation. However there is a tendency that the amount of change between the phonemes depends on the speaker. It is difficult to separate the utterance speed with a simple threshold. Therefore the sufficient accuracy with detection of speed variation was hard to be acquired.

We also attempted to acquire the correlation between dynamic measure of power or pitch with speech rate. Although there have some encouraging results showed that the measure of power is an effective way to detect the variation of speech rate, some kind of phoneme (For example, /d/ in Mandarin) will greatly increase the amount of dynamic measure. As for the F0 contour, it clearly shows the transition of the voiced and unvoiced segments. However F0 contour seems to have not enough detailed information in local speech rate variation.

As mentioned above, each of the three acoustic parameters shows some linear correlations with the speech rate. Therefore in this paper we propose a new measure for speech rate recognition, which are calculated based on the variation amount of LPC cepstrum, power and F0 contour. As ourpurpose is to find a general measure independent of specific language, both Chinese mandarin and Japanese have been used in our experiment.

2. Experimental data

Experimental data is recorded on DAT in a soundproof room. Two kinds of data are recorded. One is read by two females and one male mandarin native speaker using standard mandarin. The texts are ten sentences chosen from the Chinese magazine "China Scholars Abroad". The speaker read in two types of speech rate. (Type A: Utter quicker than normal. Type B: Utter slower than normal.) 60 sentences (10 sentences x 3 speakers x 2 types) are recorded in all.

The second kind of data is uttered by a male Japanese speaker using Japanese. The speaker uttered with local speech rate variation.

The recorded data are sampled at 12kHz/16bit.

3. Proposed measure of speech rate

In this section, we will first introduce the three groups of parameters and then discuss the method to combine them.

3.1. LPC Cepstrum

The analysis factors are shown in Table 1.

Table1. LPC cepstrum analysis factors

Window	Hanning
LPC	12 dimensions
LPC Cepstrum	20 dimensions
Frame length	20ms
Frame shift term	5ms

Next, the dynamic measure [4] based on LPC cepstrum is computed. This measure canexpress the amount of spectral change equivalent to difference smoothly.

3.2. Power and F0 contour

Power and F0 are calculated by a program get_f0 of ESPS[5]. The analysis window is hanning window. The width of an analysis frame is 30ms and consecutive frames are shift by 5ms.

The same dynamic measures are calculated with the two parameters.

Fig.1 shows the results of these three parameter's dynamic variation amount.

3.3. 3-dimension measure

Because the three parameters are of different orders of magnitude, we carried out

a normalization process before the attempt to combine these dynamic features on speech rate. The normalization coefficients are defined as the inverse of the mean of the dynamic variation amount calculated over a sentence for each analysis factor, i.e., LPC cepstrum, power, F0.

The dynamic measure for the 3-dimension parameters is defined by the following equation:

$$D = w_{cep} \sum_{i=1}^{n} a_{i\,cep}^{2} + w_{pow}\, a_{pow}^{2} + w_{pit}\, a_{pit}^{2}, \quad (1)$$

where n is the LPC cepstrum order, ω_{cep}, ω_{pow}, ω_{pit} are normalization factors for cepstrum, power and pitch, respectively. α_{icep}, α_{pow}, α_{pit} are the inclinations of the regression curve obtained by the weighted coefficients on each order.

4. Relation between average variation amount and speech rate

In order to detect the variation of local speech rate by the proposed dynamic measure, short time average processing is performed. The frame length is 2 second and the frame shift is 0.5 second. The result is shown in Fig.2. We found that the change of the curve coincides with the local speech rate variation. When the utterance goes faster, a larger amount of variation is observed, and vice versa.

5. Discussion

The proposed measure is a synthetical parameter to describe the speech rate property. Because it is comprehensively combined, it can fairly express the variations neglected by single feature, especially by F0 contour. It can also suppress the unexpected peaks often observed in the approach with power. In other word, when a reliable acoustical change happens, all of the three features provide a peak at almost the same position, so that the combined parameter will provide a large peak at that point.

In this study, we gave the three parameters the same weight, i.e., equally normalized. It seems necessary to modify the value of weight to different speech style and language. In our further study, we will attempt to obtain a sounder standard to determine the threshold used in calculating the average variation instead of an empirical value.

6. Conclusion

In this study we have originally presented a speech rate calculation method comprised of three acoustic parameters. It was shown that the new measure is consistent with the local speech rate variation. It can be applied to speech rate perception and local speech rate variation perception.

Acknowledge

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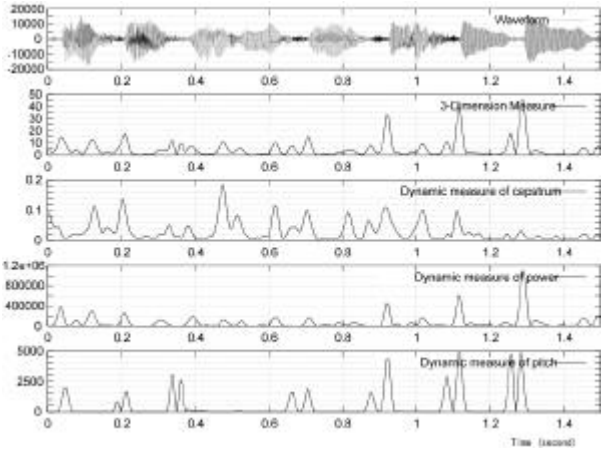


Fig. 1. 3-Dimensional Dynamic measure for detecting speed variation

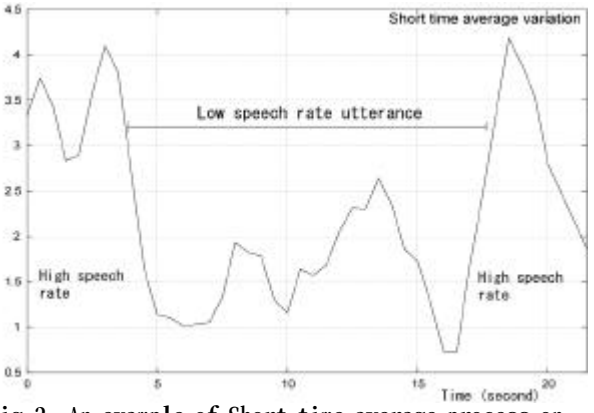


Fig.2. An example of Short time average process on an utterance with large speech rate variation